

MEININGEN, Germany

WMO: 105480

Lat: 50.5611N

Lon: 10.3772E

Elev: 453

StdP: 96.00

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.0	-11.3	-18.8	0.8	-11.7	-15.8	1.0	-11.0	12.1	4.3	10.6	3.1	2.3	20	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.2	17.6	25.5	16.9	23.7	16.2	18.9	24.9	17.7	23.3	16.8	22.1	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.5	21.0	15.6	11.7	20.0	14.7	11.0	19.3	55.2	25.0	51.6	23.2	48.8	22.2	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.4	8.1	7.0		-15.5	30.9	3.8	1.5	-18.3	31.9	-20.5	32.8	-22.6	33.6	-25.4	34.6	(4)
(5)			DB	-16.1	20.8	3.9	1.1	-18.9	21.6	-21.1	22.2	-23.3	22.8	-26.1	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-1.6	-1.8	2.7	6.2	11.2	14.0	16.4	16.3	12.7	8.0	2.5	0.1	(6)	
	DBStd	7.57	5.06	4.55	3.84	3.64	3.90	3.65	3.35	3.24	3.10	3.52	3.59	4.03	(7)	
	HDD10.0	1699	359	329	228	123	34	5	0	0	8	81	225	306	(8)	
	HDD18.3	4079	617	563	486	366	222	136	79	79	171	321	475	564	(9)	
	CDD10.0	709	0	0	1	8	71	126	199	197	88	19	0	0	(10)	
	CDD18.3	46	0	0	0	0	1	7	20	17	1	0	0	0	(11)	
	CDH23.3	441	0	0	0	0	9	65	184	168	14	0	0	0	(12)	
	CDH26.7	71	0	0	0	0	0	8	29	34	0	0	0	0	(13)	
(14) Wind	WSAvg	2.9	3.5	3.0	3.2	2.9	2.6	2.6	2.5	2.5	2.8	2.9	3.1	3.0	(14)	
(15) Precipitation	PrecAvg	650	56	42	47	38	57	65	75	60	55	50	56	60	(15)	
	PrecMax	804	101	116	106	79	160	146	149	140	125	167	100	117	(16)	
	PrecMin	489	15	5	9	2	3	12	23	11	11	8	1	10	(17)	
	PrecStd	81	23	26	26	18	34	34	32	35	26	31	22	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	11.2	16.2	19.4	24.7	27.9	29.4	30.3	25.3	19.7	11.6	10.6	(19)	
		MCWB	7.3	8.4	9.6	11.7	15.2	18.6	19.4	18.7	15.3	14.0	9.3	8.6	(20)	
	2%	DB	7.0	7.9	12.5	17.1	22.2	25.2	27.2	27.1	22.9	17.0	10.1	8.5	(21)	
		MCWB	5.6	6.3	8.1	10.5	13.9	17.1	17.8	17.2	16.0	12.7	8.5	7.0	(22)	
	5%	DB	5.7	5.6	10.0	15.1	20.5	23.2	25.5	25.1	20.5	14.8	8.7	7.1	(23)	
		MCWB	4.5	4.1	6.6	9.4	13.4	16.1	17.3	16.7	14.7	11.8	7.3	5.9	(24)	
	10%	DB	4.5	3.9	8.1	13.0	18.7	21.1	23.5	23.0	18.5	13.1	7.3	5.8	(25)	
		MCWB	3.3	2.5	5.6	8.1	12.5	15.0	16.5	15.9	13.9	10.8	6.0	4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	8.7	10.4	12.4	16.5	19.8	20.5	20.4	17.6	15.1	10.1	8.9	(27)	
		MCDB	8.7	10.8	14.5	17.9	21.7	25.9	27.7	27.9	21.4	18.4	11.0	9.9	(28)	
	2%	WB	5.8	6.4	8.5	11.0	15.0	17.9	19.0	18.4	16.3	13.3	8.7	7.2	(29)	
		MCDB	6.9	7.9	11.8	16.0	20.7	23.0	25.2	24.7	20.9	15.7	9.8	8.2	(30)	
	5%	WB	4.7	4.4	7.1	9.7	13.9	16.7	17.9	17.4	15.3	12.2	7.4	6.0	(31)	
		MCDB	5.6	5.4	9.8	14.2	19.5	21.7	23.6	23.2	19.6	14.4	8.6	7.1	(32)	
	10%	WB	3.4	2.6	5.8	8.6	12.8	15.5	16.9	16.5	14.4	11.1	6.1	4.8	(33)	
		MCDB	4.4	3.8	7.9	12.4	18.0	20.2	22.2	21.7	18.0	13.0	7.1	5.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.7	6.7	8.5	9.1	8.5	9.5	9.7	8.4	6.9	4.7	3.9	(35)	
		MCDBR	4.9	7.5	10.3	12.3	12.2	12.1	13.2	13.5	12.2	9.6	6.0	5.0	(36)	
		MCWBR	4.5	5.9	6.9	7.3	6.6	6.1	6.2	6.0	6.4	6.0	5.1	4.6	(37)	
	5% WB	MCDBR	4.8	7.3	9.5	11.2	11.3	11.1	11.9	11.6	10.5	8.4	5.7	4.9	(38)	
		MCWBR	4.6	6.1	6.7	6.9	6.6	6.1	6.1	5.8	6.0	5.7	5.0	4.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.298	0.320	0.356	0.393	0.398	0.397	0.409	0.402	0.370	0.346	0.323	0.295	(40)	
	taud		2.415	2.370	2.322	2.258	2.273	2.294	2.260	2.302	2.376	2.437	2.439	2.413	(41)	
	Ebn at Noon		741	813	842	847	858	861	842	829	819	773	700	684	(42)	
	Edh at Noon		61	82	103	123	127	125	128	117	97	77	60	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.77	1.50	2.57	4.07	4.84	5.29	5.02	4.33	3.11	1.82	0.85	0.58	(44)	
	RadStd		0.14	0.23	0.34	0.53	0.56	0.45	0.60	0.44	0.44	0.22	0.12	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+50	N/A

Nomenclature: See separate page